

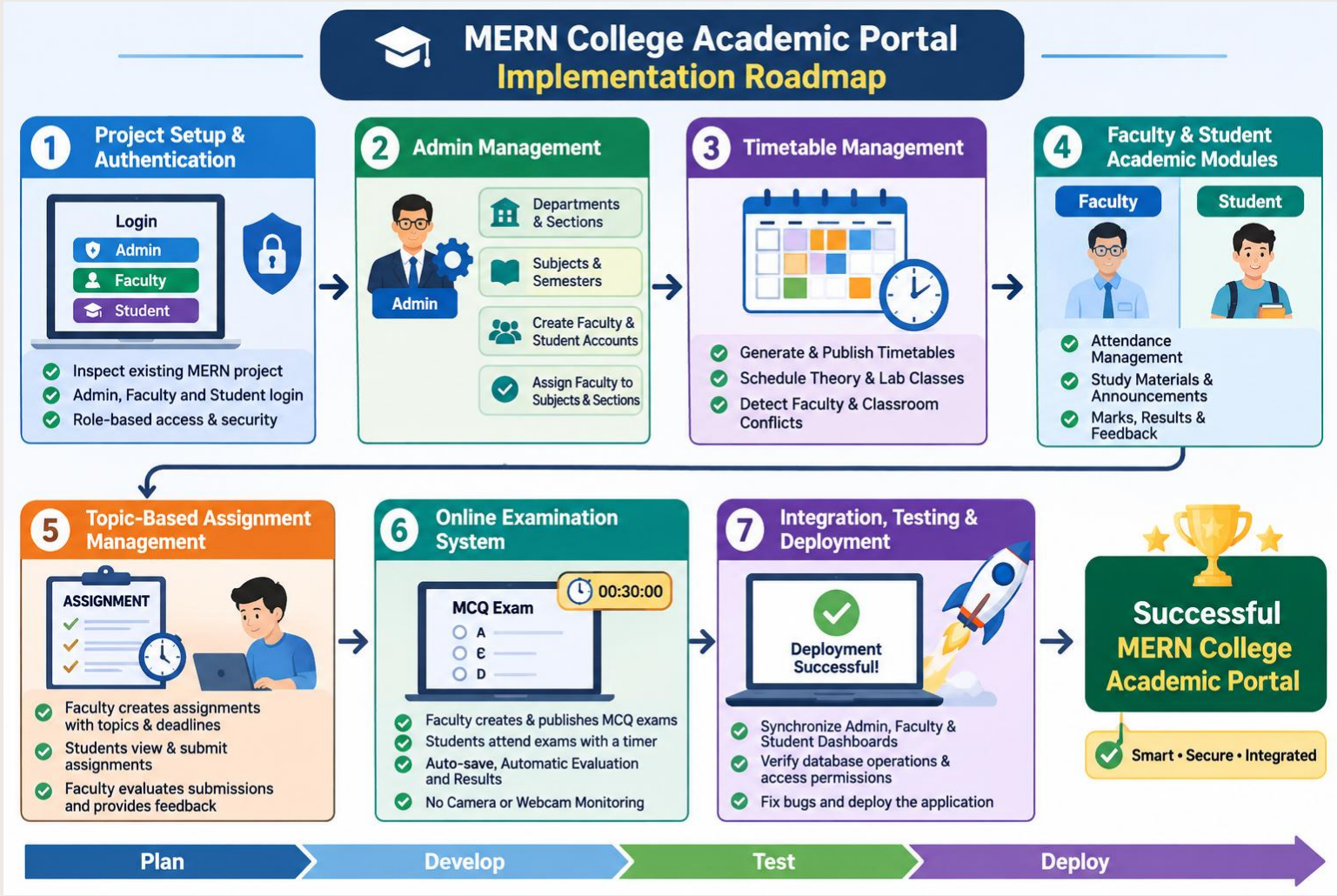
EDU TRACK -

Student Academic Management System

WHY EDU TRACK?

- **Centralized Platform:** Brings Admin, Faculty, and Students together in one system.
- **Automated Management:** Reduces manual work in academic administration.
- **Real-Time Updates:** Keeps students and faculty updated with the latest academic information.
- **Smart Timetable:** Provides organized, section-wise and faculty-wise schedules.
- **Performance Analysis:** Helps identify students' academic strengths and weak areas.
- **Improved Coordination:** Connects Admin, Faculty, and Students efficiently.
- **Digital Learning Resources:** Provides easy access to study materials and announcements.
- **Time Efficiency:** Saves time and reduces paperwork.

IMPLEMENTATION ROAD MAP



ROLES AND RESPONSIBILITIES:

ADMIN ROLE

- Manages departments, sections, and subjects.
- Creates and manages faculty and student accounts.
- Assigns faculty to subjects and sections.
- Generates and manages class timetables.
- Monitors overall academic activities.

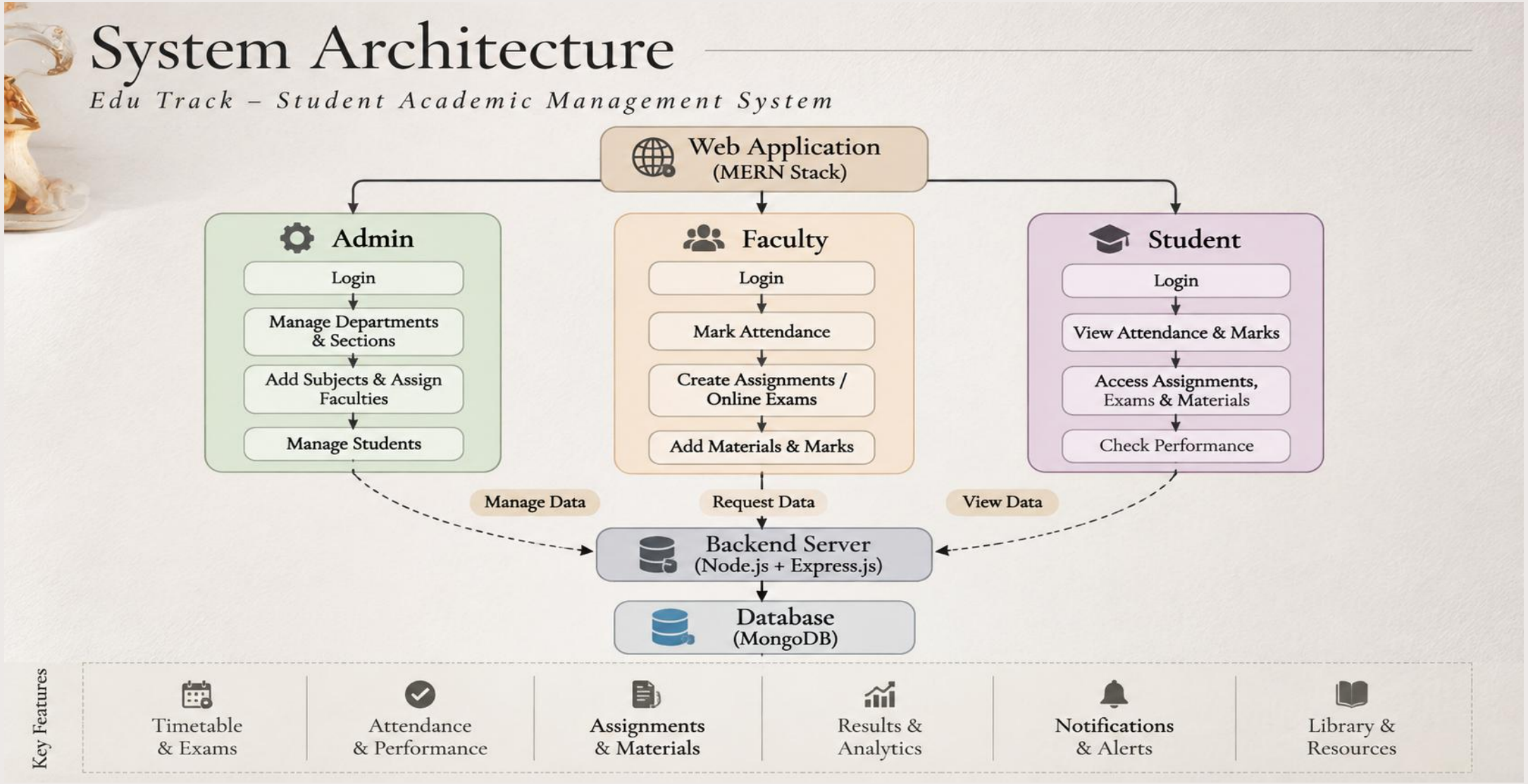
FACULTY ROLE

- Views assigned subjects, sections, and students.
- Marks and manages student attendance.
- Creates assignments and evaluates submissions.
- Conducts online examinations and publishes results.
- Uploads study materials and announcements.
- Monitors student academic performance.

STUDENT ROLE

- Views personal profile, subjects, and timetable.
- Checks attendance and attendance percentage.
- Views and submits assignments.
- Attends online examinations.
- Accesses study materials and announcements.
- Views marks, results, and faculty feedback.

SYSTEM ARCHITECTURE:



DATABASE COLLECTIONS:

User Collection – Stores login credentials and roles of Admin, Faculty, and Students.

Department Collection – Stores department and section details.

Subject Collection – Stores subject names, codes, semesters, and credits.

Faculty & Student Collection – Stores faculty and student profiles and their academic details.

Attendance Collection – Stores student attendance records.

Assignment & Examination Collection – Stores assignments, online exams, questions, submissions, and results.

Timetable Collection – Stores class schedules, subjects, faculty, and section details.

TECHNOLOGIES USED:

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Frontend: React.js, HTML, CSS, JavaScript

Backend: Node.js, Express.js

Database: MongoDB

Authentication: JSON Web Token (JWT)

Password Security: bcrypt

Additional Tools: Socket.IO, REST APIs

Development Tool: Visual Studio Code / Antigravity

ADVANTAGES:

- **Saves Time:** Reduces manual academic record management and paperwork.
- **Centralized Management:** Manages departments, faculty, students, and subjects in one system.
- **Easy Access:** Students can view attendance, timetable, assignments, and results online.
- **Better Communication:** Enables faculty to share announcements, materials, and feedback with students.
- **Online Assessment:** Supports online examinations, assignments, and automatic MCQ evaluation.
- **Secure Access:** Provides role-based login and access for Admin, Faculty, and Students.

CONCLUSION:

- **Centralized Platform:** Edu Track brings Admin, Faculty, and Students together in one digital academic management system.
- **Simplified Administration:** It organizes departments, sections, subjects, faculty details, and student records in one place.
- **Faculty Support:** Faculty can manage attendance, create assignments, share learning materials, conduct online MCQ examinations, and monitor student performance.
- **Student Access:** Students can view their timetable, attendance, assignments, examinations, results, materials, and academic updates through their own dashboard.
- **Organized and Secure Access:** The MERN-based application uses role-based access so each user can work with the features and information relevant to their role.
- **Overall Impact:** Edu Track aims to reduce paperwork, save time, improve communication, and make academic activities easier to manage.

THANK YOU

BY SYSTEM SAVIOURS